MACHINE LEARNING PROJECT

Travel Time Prediction

Using Ridge and Lasso Regression

PRESENTED BY

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Abstract

Travel Time Prediction Using Ridge and Lasso Regression

PROJECT OVERVIEW

Accurate travel time prediction is a critical component of intelligent transportation systems, enabling efficient route planning, traffic management, and informed decision-making for commuters. Travel time is influenced by travel distance, traffic conditions, time of travel, road characteristics, and other environmental and operational parameters.

The proposed system employs Ridge Regression and Lasso Regression. The dataset is preprocessed through missing-value handling, inconsistency removal, feature selection/transformation, train-test splitting, and feature scaling.

Ridge applies L2 regularisation, while Lasso applies L1 regularisation and can perform implicit feature selection. Both models are evaluated using MAE, MSE, RMSE, and R^2.

Problem Statement

Travel Time Prediction in Intelligent Transportation Systems

Accurately predicting travel time is a key problem in Intelligent Transportation Systems. Traditional estimation methods struggle to capture
how multiple factors interact, creating the need for a machine learning-based
predictive system that learns patterns from historical travel data.

PROJECT OBJECTIVE

Develop a supervised machine learning model that predicts travel time using relevant travel-related features, reduces overfitting through regularisation, and compares Ridge and Lasso Regression to identify the better-performing model.

FACTORS INFLUENCING TRAVEL TIME

1. Travel Distance
2. Traffic Conditions
3. Time of Travel
4. Road Characteristics
5. Other Operational Factors

System Architecture



ALGORITHM 1

Ridge Regression

L2 Regularisation

$$\text{Ridge Loss} = \Sigma(y - \hat{y})^2 + \lambda\Sigma\beta^2$$

Adds an L2 penalty on model coefficients, shrinking them and improving stability.

How Ridge Works

Ridge Regression for stable travel-time prediction

1. Learns the relationship between travel features and travel time
2. Calculates prediction error
3. Adds a penalty for very large coefficients
4. Coefficients are reduced / shrunk
5. Model becomes more stable and less sensitive to correlated features

WHY RIDGE IS USEFUL

Travel-related features may be correlated. Ridge reduces multicollinearity and prevents excessively large coefficients, improving stability and generalisation.

ALGORITHM 2

Lasso Regression

L1 Regularisation

$$\text{Lasso Loss} = \Sigma(y - \hat{y})^2 + \lambda\Sigma|\beta|$$

Uses an L1 penalty. Less important coefficients can become zero, enabling feature selection.

How Lasso Works

Lasso Regression for simpler predictive models

1. Starts with available travel features
2. Learns their relationship with travel time
3. Calculates prediction error
4. Adds an L1 penalty
5. Less important coefficients can become exactly zero
6. Zero-coefficient features are effectively removed

WHY LASSO IS USEFUL

Lasso can automatically perform feature selection and produce a simpler model containing the most useful predictors.

Algorithms Used & Evaluation

Common workflow followed by Ridge and Lasso

1. PREPARE THE DATA

Handle missing values
Remove inconsistencies
Transform variables
Select relevant features

2. SCALE & SPLIT

Feature scaling
Training data → model learning
Testing data → evaluation

3. TRAIN MODELS

Ridge Regression (L2)
Lasso Regression (L1)
Both trained on prepared data

4. PREDICT

Both models receive test features and produce predicted travel durations.

5. EVALUATE & COMPARE

MAE • MSE • RMSE • R^2
Select the model with better accuracy and generalisation.

Dataset Identified

Travel Time Prediction Dataset

FORMAT

CSV (.csv)

DATA TYPE
Historical travel-related data

PROBLEM TYPE
Supervised Learning → Regression

TARGET VARIABLE
Travel Time

INPUT FEATURES

- Travel Distance
- Traffic Conditions
- Time of Travel
- Road Characteristics
- Other Operational Parameters

TARGET: TRAVEL TIME

Note: The project document identifies the dataset as a travel time CSV dataset and does not specify a particular source or filename.

Model Comparison

Ridge vs Lasso for Travel Time Prediction

RIDGE REGRESSION

REGULARISATION
L2

PENALTY
β^2

COEFFICIENTS
Shrinks coefficients

FEATURE SELECTION
No direct selection

MAIN OBJECTIVE
Stability + generalisation

LASSO REGRESSION

REGULARISATION
L1

PENALTY
$|\beta|$

COEFFICIENTS
Can make coefficients zero

FEATURE SELECTION
Yes

MAIN OBJECTIVE
Simplicity + feature selection

FINAL MODEL SELECTION

Compare MAE, MSE, RMSE and R^2 → select the model with better predictive accuracy and generalisation.

Thank You

Travel Time Prediction Using Ridge and Lasso Regression

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